ANALYSIS OF GENE FLOW BETWEEN GEOGRAPHICALLY SEPARATED POPULATIONS OF *ETHEOSTOMA RADIOSUM* IN NORTHEASTERN TEXAS AND SOUTHEASTERN OKLAHOMA

A Thesis

by

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ABSTRACT

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Victoria E. Hardeman, MS
Texas A&M University-Commerce, 2023

Advisor: Bjorn Schmidt, PhD

The genus Etheostoma has been found to be one of the most cryptic groups in North America today. Many darter species are habitat specialists that have limited dispersal in drainages across unsuitable conditions. *Etheostoma radiosum* is found in the Red River system of Oklahoma, Arkansas, and Texas, where it is more common in upland tributaries with gravel, indicating that the muddier main stem of the Red River may restrict gene flow patterns between systems. The purpose of this study is to use molecular markers to determine genetic population structure of this species in Texas populations and potential barriers and genetic isolation. We sampled five populations in Texas tributaries and three populations in Oklahoma in different tributary systems in 2022. Captured individuals had caudal fin clips taken in the field, followed by DNA extraction and amplification at six microsatellite loci. Pairwise F_{ST} values, observed and expected heterozygosity, STRUCTURE analyses, isolation by distance, and discriminant

analysis of principal components were done to determine population genetic patterns in the region. Results indicate the existence of three distinct genetic clusters in the region.

ACKNOWLEDGEMENTS

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Lastly, I would like to mention my parents, Dinah and Gary Hardeman, and my best friend and partner, Matthew Swerdfeger, for being with me through all of this. The inspiration and motivation they have given has kept me working through the most difficult times.

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Chapter 1

INTRODUCTION

Etheostoma radiosum (orange-bellied darter) have very specific habitat requirements. Specifically, they prefer clear water with a cobble substrate (Page and Burr 1991). Their ecology requires riffles in running streams in order to survive and thrive. Furthermore, they use the cobble substrate to bury their eggs and keep them safe from predators (Scalet 1973). The quick flowing water keeps the eggs from becoming infested with fungi and facilitates survival of the eggs and larvae. *Etheostoma radiosum* are found in tributaries of the Red River in Arkansas, Oklahoma, and Texas, specifically, where the water is clearest and quickest moving. Unfortunately, this habitat has become rather uncommon in tributaries near the main stem of the Red River in Texas, where it is listed by Texas Parks and Wildlife (TPWD) as species of greatest conservation need because of its rarity in the state.

This study examines populations occurring in Texas along tributaries of the middle Red River between Lake Texoma and the Texarkana as well as populations occurring in the upland tributaries of the Red River in Oklahoma. These areas were chosen because the tributaries along the south side of the Red River are typically quite different from what this species requires. Most of the stream sections are wide and consist of slow-moving, and sometimes stagnant, water. Additionally, the substrate is typically very silty and muddy with increasingly high turbidity. This is the result of most of the land around the Red River being used for agricultural or urban development. This means that the species has been pressured to move up the branches to specific locations in the region to find a more suitable habitat or has become extirpated in the region. Although extant populations are still found in recent collections, populations may be separated

along the main stem because of lack of suitable habitat in adjacent tributaries (Birdsong et al. 2020).

One of the main goals of this study was to determine if each major drainage along the Red River constituted individual populations of the species. The major drainage systems in Oklahoma relevant to our study area that historically have had *E. radiosum* are the Washita River, Blue River, Muddy Boggy River, and Kiamichi River. Historically, the Blue River was thought to be the western edge of the species' range (Scalet 1973), but population were found in the Washita River system in the 1980s (Matthews and Gelwick 1988). Further analysis showed that the Blue River had a distinct morphology, and was elevated to a distinct species, *E. cyanorum* in 2019 (Matthews and Turner 2019). The Washita drainage west of the Blue River matched the *E. radiosum paludosum* subspecies in the Muddy Boggy and Kiamichi, introducing an apparent disjunction of the range in the middle Red (Matthews and Gelwick 1988, Matthews and Turner 2019). Because of the disjunction around the Blue River, and the avoidance of muddy habitat near the main stem, it has long been speculated that the main stem acts as a barrier to gene flow, isolating populations within the major tributaries (Scalet 1973, Matthews and Gelwick 1988). Specifically we wanted to test if the populations that were located along the main stem of the Red River would show genetic signs of isolation, with each main system (Washita, Muddy Boggy, and Kiamichi) being genetically distinct. If a population was able to speciate entirely from the other lineages in the area, there is potential reason to believe further speciation (or at least genetic isolation) may have occurred in other drainages of the Red River. We also wanted to see if populations near the main stem of the Red in Texas were genetically related to populations in the main tributary systems in Oklahoma. The work done by Matthews and Turner (2019) demonstrates that there is significant fragmentation occurring in the habitat

for this species. This study aims to determine where those distinct breaks in gene flow are occurring along the Red River and be able to provide information for what those barriers may be.

Isolation of a species within the Blue River with no movement into the main stem of the Red River suggests that the species is likely unable to use the intermittent habitat occurring between the upland stretches of the Blue River and the main stem of the Red River. This would make sense because the lower areas of this drainage are highly degraded stream sections with high amounts of mud and silt as the primary substrate. Additionally, this barrier effect of the muddy main stem and connected smaller systems could be a primary cause for differentiated populations along the other drainages we sample. The Washita populations which were discovered in the 1980s also face other potential barriers to dispersal, as in Oklahoma these populations occur upstream of Lake Texoma (Washita is the northern branch of the lake). Lake Texoma was constructed in 1944 and likely introduced more fragmentation and isolation for these populations. Because of the effects of reservoir fragmentation, we expect to see more genetic clusters to the west of the Blue River and more continuous gene flow present in the eastern parts of our study area.

Chapter 2

REVIEW OF THE LITERATURE

Recent work of Matthews and Turner (2019) suggests a disjunction between the major tributaries of the Red River that would mean an unusual and non-continuous spatial pattern for the species. Matthews and Turner (2019) were able to describe a completely new species from the populations in this area. *E. cyanorum* is only found in the Blue River and is considered a completely separate species from *E. radiosum*. In addition, *E. radiosum* population further east in the Little and Ouachita Rivers in Oklahoma and Arkansas (still tributaries of the Red River) are a separate subspecies than the populations in the Kiamichi, Muddy Boggy, and Washita systems (Matthews and Turner 2019). This isolation of different groups in different tributary systems of the Red highlights how the muddy main stem sections act as a barrier for gene flow. However, with regard to *E. cyanorum* and the proposed study range, it seems unlikely that the populations to either side of the newly defined species would retain enough gene flow to still constitute their description as the same species. If the metapopulation has become so fragmented as to necessitate the need for the redescription of a subspecies as a new species, then all subspecies should be adequately analyzed to determine if any more speciation has occurred.

When a population of a species is completely separated from other populations there is a lack of crossbreeding and gene flow. This leads to genetically distinct populations over time (Mendelson et al. 2007) and speciation, a phenomenon caused by genetic isolation. If the isolation continues, then eventually the populations become so genetically distinct that they are unable to breed with other populations. This means that they are no longer the same species according to the biological species concept (Rundle and Nosil 2005). A study by April et al. (2011) stated that within the genus of *Etheostoma* there is a large amount of cryptic diversity in

many genomes that should be explored, including within the current range of *E. radiosum*, which encompasses three distinct lineages. Also, early allozyme analysis indicated high genetic differentiation within tributary systems of Oklahoma for *E. radiosum* (Echelle et al. 1975). Both of these results indicate that the often turbid and saline main stem of the Red River may act as a barrier to dispersal, restricting populations of *E. radiosum* in tributaries and causing genetic isolation.

It is important to understand why a species has become genetically isolated for numerous reasons. As with the Fountain Darter (*Etheostoma fonticola*), lack of gene flow can cause a species to become endangered, threatened, or even completely extirpated (Olsen et al. 2016). This is due to an increased risk of bottleneck events or inbreeding depression caused by a greatly decreased number of populations connected by dispersal. Knowing and understanding these differences between populations means that the best possible means for conservation and preservation of genetic diversity and promote sufficient gene flow to maintain healthy metapopulations can be determined. This information could be used to propose genetic rescue by using this information on the genomes of different populations to determine which ones could benefit from increased gene flow. Then work could begin intervening by manually moving individual fish from one population to another during or before breeding season if necessary. As E. radiosum is listed as a species of greatest conservation need in the state of Texas, an assessment of the genetic health and lineages of populations could be helpful for guiding any state-level conservation efforts.

Species of *Etheostoma* have specifically been very successful in their spread throughout many different ecosystems and habitats of North America (Fluker 2011). Their radiation over North America has presented a challenge when it comes to adequately tracking the number of

species, as cryptic diversity is common in the group. As their populations have become increasingly dispersed, they have also become increasingly diverse, and their collective genome encompasses over 140 distinct populations (Kuehne and Barbour 2015). However, as their environments are continuously altered, and barriers to gene flow are created, it is important to pay attention to the amount of gene flow that is occurring in this group because of their high rate of natural fragmentation and propensity for habitat specialization. Understanding how populations of *E. radiosum* are being affected and changed at the genetic level gives better insight into how to best study and understand their needs and provides insights to a complicated biogeographic pattern for the species group.

In this study, I describe the genetic composition of the known populations of *E. radiosum* in Texas tributaries of the Red River basin in relation to major tributaries in Oklahoma. They have historically been located in at least two different places in Texas but were able to be found in five Texas sites in 2022. For comparison with Oklahoma patterns, three sites were sampled (one each from the Washita, Muddy Boggy, and Kiamichi). Little is known about the amount of gene flow taking place between these populations and smaller tributaries adjacent to the Red, and this presents an opportunity for research. The results of this study will help managers to understand what impacts development and fragmentation has had on the study species. The main obstacle I anticipated with this research was getting enough samples from locations within Texas. The samples I was able to get are assumed to be a good measure of the Texas population as a whole, as 34 sites were surveyed along the Red, leading to collection of the species at these five locations.

Chapter 3

MATERIALS AND METHODS

Study Sites

The sites where *E. radiosum* have been found historically are located in prairie and grasslands. The water is somewhat clear with abundant riffles throughout. The streambed of occurrence sites is typically composed of cobble and gravel substrate. Average precipitation in this area is from 114-140 centimeters annually (Newcomer et al. 2017). The soil composition of this region is highly acidic with a loamy soil that is a high proportion of sand. The temperature is typical of the area in northeastern Texas and has an annual high of 22°C and an average low of 11°C (USGS n. d.). The area is generally flat, with some slow-sloping hills.

In addition to the sites where they have been located historically, similar sites close to these were also sampled. In total I was able to capture 220 individual fish from eight different sites. Five of the sites were Texas locations and three Oklahoma sites were also used (Figure 1, Table 1). A total of three sites were located above an established dam (TX04, TX05, and OK01) and the rest were located with eventual downstream access to the Red River. Furthermore, each site was a wadable stream that is accessible for public use.

Table 1: Each population that was sampled, and the number of individuals caught at each site that were then used for genetic analyses.

Site name	Number of individuals analyzed	Locality
TX01	30	Lower Shawnee Creek
TX02	30	Duck Creek
TX03	13	Sanders Creek
TX04	28	Little Mineral Creek
TX05	30	Upper Shawnee Creek
OK01	29	Glasses Creek
OK02	30	Rock Creek
OK03	30	Sandy Creek

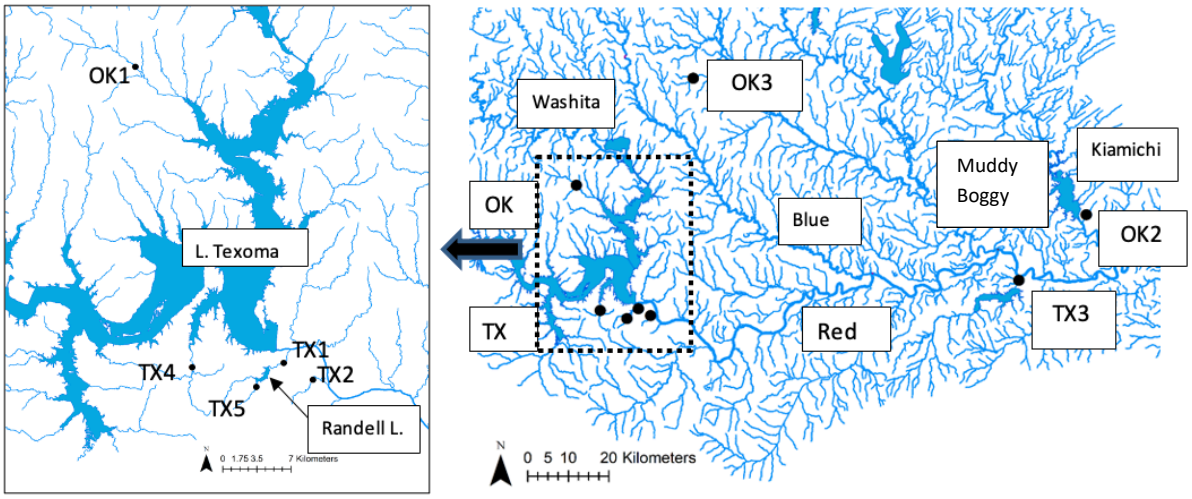

Figure 1: A map of the sites where samples were taken. The inlay shows a close up of the indicated area so these sites can be seen more accurately.

Trapping

Efforts to capture *E. radiosum* were focused on sites where they were found historically and similar sites nearby these locations. The original goal was to locate ten Texas populations of the species. However, this proved to be more of a challenge, so I chose to take the data from the five Texas sites we found and compare them to the three populations I was able to sample in Oklahoma. The hope was that this could provide more information about the amount of movement taking place along the Red River and how it is different and similar to what is occurring in the upland tributaries of Oklahoma.

At each site my team and I used an electrofishing backpack unit, (Aqua Shock Solutions LLC, Sevierville, Tennessee) to capture fish. These fish were then sorted and identified to find *E. radiosum*. Our target sample size was at least 30 and no more than 50 individuals from each site. Assessments into the number of fish caught were conducted at the end of a 150-meter stretch. A caudal fin clip was taken from each of the target fish captured and preserved in an ethanol solution until they were able to be genetically analyzed in the laboratory. All samples were then released back into the water body. Some sites had lower abundance, and in this case all individuals collected after exhaustive sampling had caudal fin clips taken for analysis (TX03).

Genetic Analysis

The fin clips were brought back to the lab, where my team and I extracted DNA samples to be used with microsatellite DNA markers to determine how similar their genetic composition is to the other populations of *E. radiosum*. Genomic DNA was extracted from tissue samples via spin columns using E.Z.N.A. Tissue DNA extraction kits (Omega Bio-Tek, Norcross, Georgia). The following loci were used as the mark for variability between populations: Eca37, Eca49, Eca70, Esc18, and Esc26. These were used by Tonnis (2006) and Gabel et al. (2008) on other

Etheostoma species to (*Etheostoma caeruleum* and *Etheostoma scotii*). PCR reactions were carried out in 12.5 µl mixtures containing 6.25 µl of OneTaq PCR master mix, 0.5 µl forward primer, 0.5 µl reverse primer, 0.25 µl template DNA, and 5µl nuclease free water. PCR cycling conditions followed recommended settings for the master mix, consisting of an initial denaturing step at 95°C for 2 mins, followed by 35 cycles of 56°C for 30s and 72°C for 1 min, followed by a final extension at 72°C for 2 min. PCR products were separated by electrophoresis on 6% polyacrylamide gels, followed by band visualization using standard silver staining protocols. Allele scoring was performed on gel image pictures using a UVP GelStudio plus.

Statistical Analysis

Microsatellite markers used were examined for any deviations from linkage disequilibrium and Hardy Weinberg Equilibrium. Additionally, for each population traditional genetic measurements were calculated including mean number of alleles, allelic richness, observed heterozygosity, and expected heterozygosity, and inbreeding coefficient (F_{IS}) using GenAlEx v 6.5 (Peakall and Smouse 2012) and PopGenReport in R (Adamack and Gruber 2014).

Pairwise F_{ST} values was calculated for all population pairs in GenAlEx to compare pairwise linearized F_{ST} ($F_{ST}/(1-F_{ST})$) as used in Rousset (1997). Isolation by distance (IBD) was assessed with linear regression of pairwise linearized genetic distance with pairwise river distance measured in ArcGIS. IBD analysis allows us to see how much genetic variation there is correlated to the geographic distance of the populations. The program Structure (Pritchard et al. 2000) was used to assess the number of distinct genetic clusters in the dataset. To complete the analyses using Structure I followed the steps outlined in the manual as well as those described by Evanno et al. (2005) to select the optimum k value. This program helps to identify how many

genetic clusters were sampled. This means that even if some E. radiosum were found in close proximity to each other, I can now identify if they are part of the same genetic population or if something is interfering with gene flow between them. A two-sample t-test was carried out in R to compare mean genetic diversity (mean expected heterozygosity) occurring in sites above and below the dam. Finally, the clusters calculated using structure were using in adegenet in R to calculate the discriminant analysis of principal components (DAPC). This gives insight into how genetically similar the different identified clusters are to one another.

Chapter 4

RESULTS

Genetic Summary

The number of alleles, allelic richness, and expected heterozygosity found in each population gives information on the diversity occurring in each population. All populations were found to be in Hardy-Weinberg equilibrium. Notably, the lowest number of alleles is found in Texas population 4 and Oklahoma population 1, both at around 3 alleles (Table 2). These also are both sites located above dams. This table also shows us that throughout all of the populations sampled our inbreeding levels were typically moderate. The inbreeding values are moderate for Texas sites 1 and 4 as well as Oklahoma site 1. On average, the sites above dams had higher inbreeding with the average value being 0.11 whereas the sites more closely connected to the main stem of the Red River had an average value of 0.052. Allelic richness (A_R) standardizes the average number of alleles across loci by sample size at that location, allowing for better comparison of genetic diversity across sites with unequal sampling. The average A_R for sites above dams is 2.84 whereas the average for sites connected to the main stem of the Red River is 3.76 indicating an increase in allelic richness in sites with a connection to the main stem.

Table 2: Summary statistics of population genetic patterns at each site. N refers to the population size, N_a refers to the number of different alleles present in each population, A_R refers to the allelic richness within each population, H_O is the observed heterozygosity for each population, H_e is the expected heterozygosity calculated, and F_{IS} is the fixation index or amount of inbreeding present in the given population.

Site name	**N**	**N_a**	**A_R**	**H_O**	**H_e**	**F_{IS}**
TX01	30	3.8	3.36	0.487	0.601	0.210
TX02	30	3.6	3.43	0.527	0.572	0.061
TX03	13	3.6	3.47	0.641	0.586	-0.094
TX04	28	2.8	2.79	0.436	0.515	0.154
TX05	30	3.4	3.14	0.480	0.512	0.017
OK01	29	2.8	2.60	0.297	0.383	0.162
OK02	30	5.8	4.77	0.677	0.694	0.031
OK03	29	4	3.74	0.550	0.582	0.065

Isolation by Distance

Sites sampled ranged from pairwise river distances of 4.73 kilometers to 322.29 kilometers. The average distance was 152.68 kilometers, 95% CI [110.83 km, 194.53 km]. Pairwise F_{ST} values ranged from 0.023 to 0.155 as show in Table 3. The average pairwise F_{ST} value was 0.079, 95% CI [0.063, 0.095]. A regression of pairwise river distance on pairwise genetic distance was completed to assess if isolation by distance patterns were present (Figure 2). Results gathered indicated that there was no significant IBD patterns and the slope of the relationship between two variable was not significantly different from zero ($R^2 = 0.016$; $p = 0.522$).

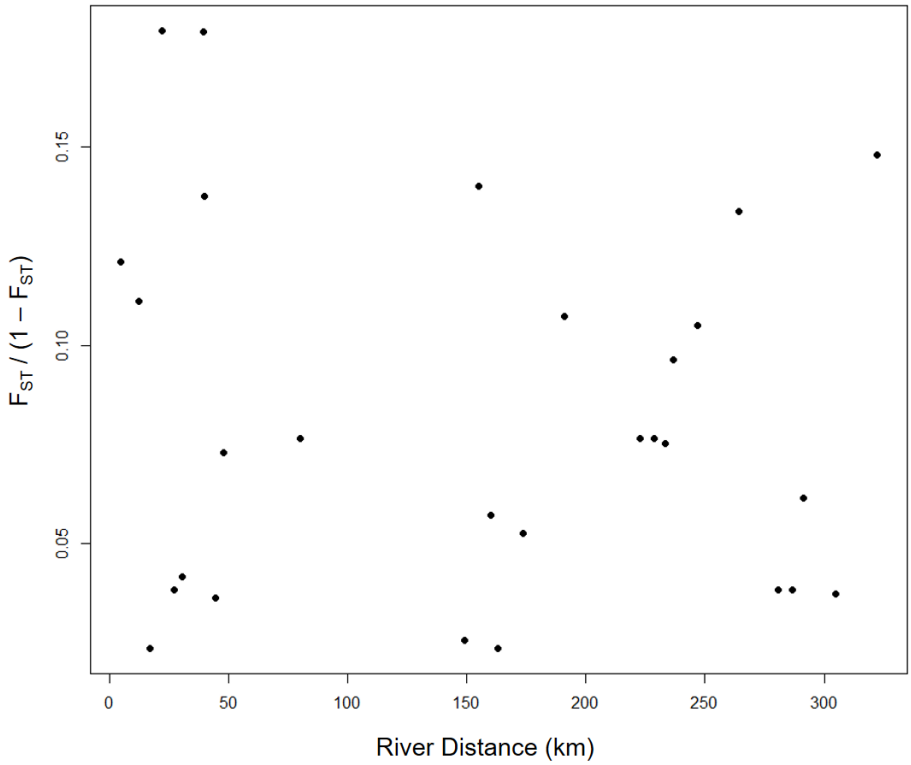

Figure 2: The F_{ST} values plotted against the distance between rivers in kilometers. Demonstrates no adhesion to expected patterns of isolation by distance.

Effects of Dams on Genetic Diversity

The program R was used to complete a two-sample t-test to determine the average expected heterozygosity above and below the dams (Figure 3). The average expected heterozygosity was 0.47 in fragmented sites, or those sites located above a dam and 0.607 in sites that are connected to the Red River, which is a significant difference as calculated with a two sample t-test ($p = 0.019$) (Figure 3). Heterozygosity is a means for measuring genetic diversity because it implies the presence of two different alleles in a single individual. There is a significant decline in genetic diversity above the dams as opposed to below the dams. This implies a lack of genetic diversity is caused by the presence of these dams and is having an

impact on the species ability to maintain steady gene flow. This is further enforced by patterns shown in pairwise F_{ST} values (Table 3). These values demonstrate how genetically dissimilar the populations are. There is a pattern of larger pairwise F_{ST} values between the sites above and below the dam. For instance, OK01 is located above a dam and shows higher pairwise values with other sites across/below the reservoir, such as a value of 0.118 with OK02.

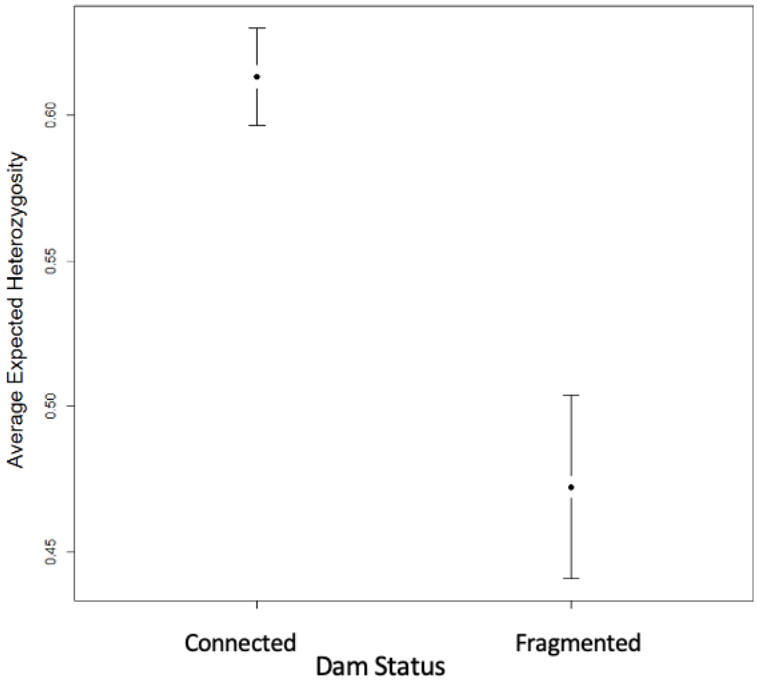

Figure 3: The average expected heterozygosity of sites that are connected to the main stem and those fragmented by the presence of a dam. The error bars shown are standard error in each group.

Table 3: The calculated F_{ST} values for each pairwise population set. The values represent how dissimilar each population is from the on it is being compared to. Highlighted values correlate to the sites located above a dam.

	TX01	TX02	TX03	TX04	TX05	OK01	OK02	OK03
TX01	0.000							
TX02	0.100	0.000						
TX03	0.123	0.025	0.000					
TX04	0.153	0.040	0.050	0.000				
TX05	0.108	0.023	0.054	0.037	0.000			
OK01	0.152	0.068	0.097	0.121	0.035	0.000		
OK02	0.071	0.071	0.071	0.095	0.070	0.118	0.000	
OK03	0.155	0.037	0.023	0.036	0.058	0.129	0.088	0.000

Structure Pattern

Analyses done with Structure demonstrated that the best supported number of distinct genetic clusters in the data was three. This was determined using the Evanno method which determines the highest likelihood with the lowest number of clusters possible (Evanno and Goudet 2005) (Figure 4). These groupings are generalizations made with the genetic components present (Figure 5). This means that there are signals from all three groupings within each site sampled but the site will be assigned to a population based on which group they had stronger signals of. Two of these groups are located west of the Blue River and includes sites in the Washita region and adjacent to Lake Texoma (Figure 6). The other group is located to the east of the Blue River and includes sites in the Muddy Boggy River, Kiamichi River, and Sanders Creek regions (Figure 6). Population one that is located in the western part of our search area and is

mostly above dams will be referred to as the west fragmented population. The other population to the west that has connection to the main stem of the Red River will be referred to as the west connected population. Finally, the population in the eastern area of our study will be referred to as the east connected population. Most of the sites had a clear majority of which cluster they were inferred to be in as shown in Table 4. However, Texas site 2, Duck Creek, demonstrated a mixed genetic population, with split percentages of 36% signal from the west connected population, 28% signal from the east connected population, and 36% signal from the west fragmented population. All other populations had between 61%-81% belonging to a single genetic cluster making their assignment more obvious. This make TX02 very unique in its genetic patterns as opposed to the other sites we sampled. The inferred clusters are signified by a number 1-3 in Table 4 and a color in Figures 5 and 6. Cluster 1 correlates to the dark purple color on the figure (called “west connected” as it has a stronger signature in western sites connected to the Red River), cluster 2 correlates to the orange color on the figure (called “east connected” as it has a stronger signature in eastern sies connected to the Red River), and cluster 3 correlates to the light blue color on the figure (called “west fragmented” as it has a stronger signature in western sites above the two reservoirs in the study).

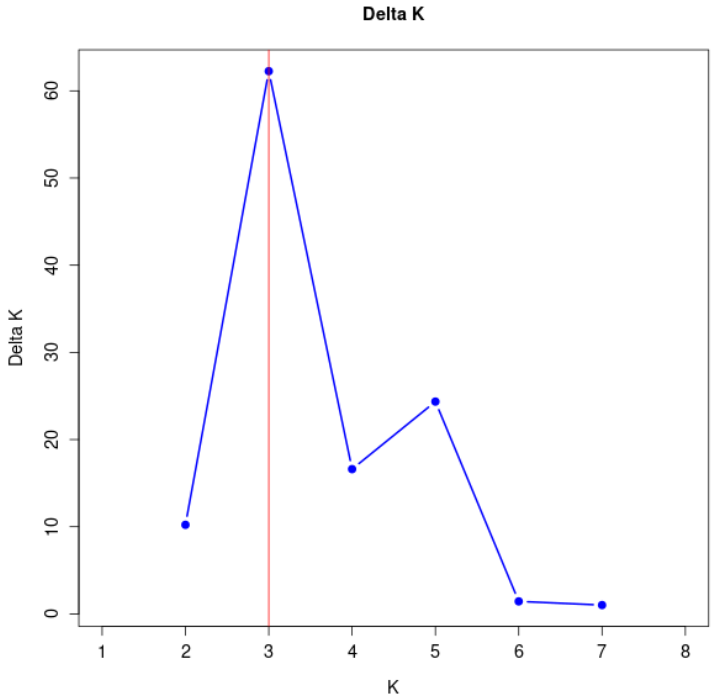

Figure 4: Evanno plot for the highest likelihood for number of clusters with the fewest number of clusters.

Table 4: Average Q scores for cluster assignment per population from Structure analysis.

	Inferred cluster		
Site name	1	2	3
TX01	0.807	0.059	0.134
TX02	0.358	0.279	0.363
TX03	0.194	0.690	0.116
TX04	0.133	0.156	0.711
TX05	0.221	0.133	0.646
OK01	0.311	0.081	0.608
OK02	0.142	0.725	0.132
OK03	0.072	0.720	0.208

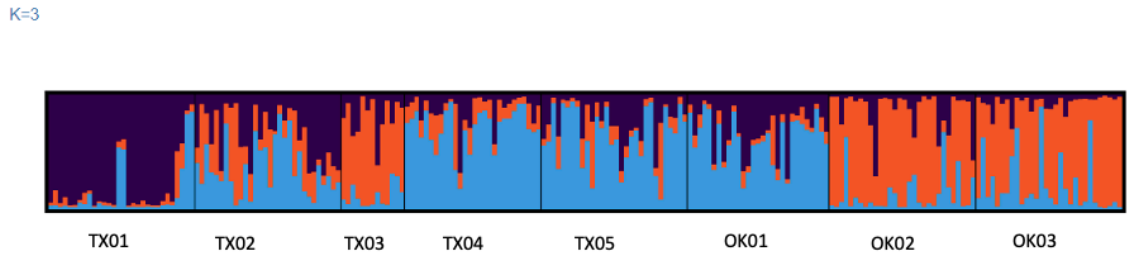

Figure 5: Structure barplot. Values shown in Table 1 are assigned a color and gives a visual representation of which cluster each individual fish has signals from. Each color is a separate cluster.

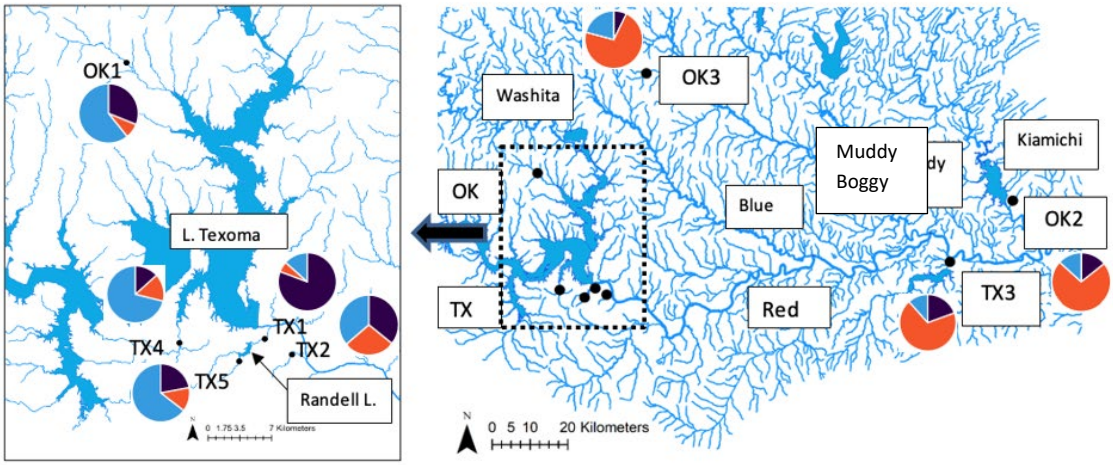

Figure 6: Map of the study sites with accompanying pie charts to show what percentage of the fish caught fit into each genetic population.

An additional analysis was done to determine how these three genetic clusters relate to each other. A discriminant analysis of principal components was done in the program R using adegenet to assess these relationships. The same groupings that were established during the Structure analysis were used in this analysis. Group 1 corresponds to the west fragmented population, group 2 corresponds to the east connected population, and group 3 corresponds to the west connected population as specified. Figure 7 shows that the east connected population and the connected west population are much more closely related than the west fragmented population. This implies that even though the two west populations are geographically closer to each other than to the eastern population, the lack of movement through the main stem of the Red River has made them less genetically similar.

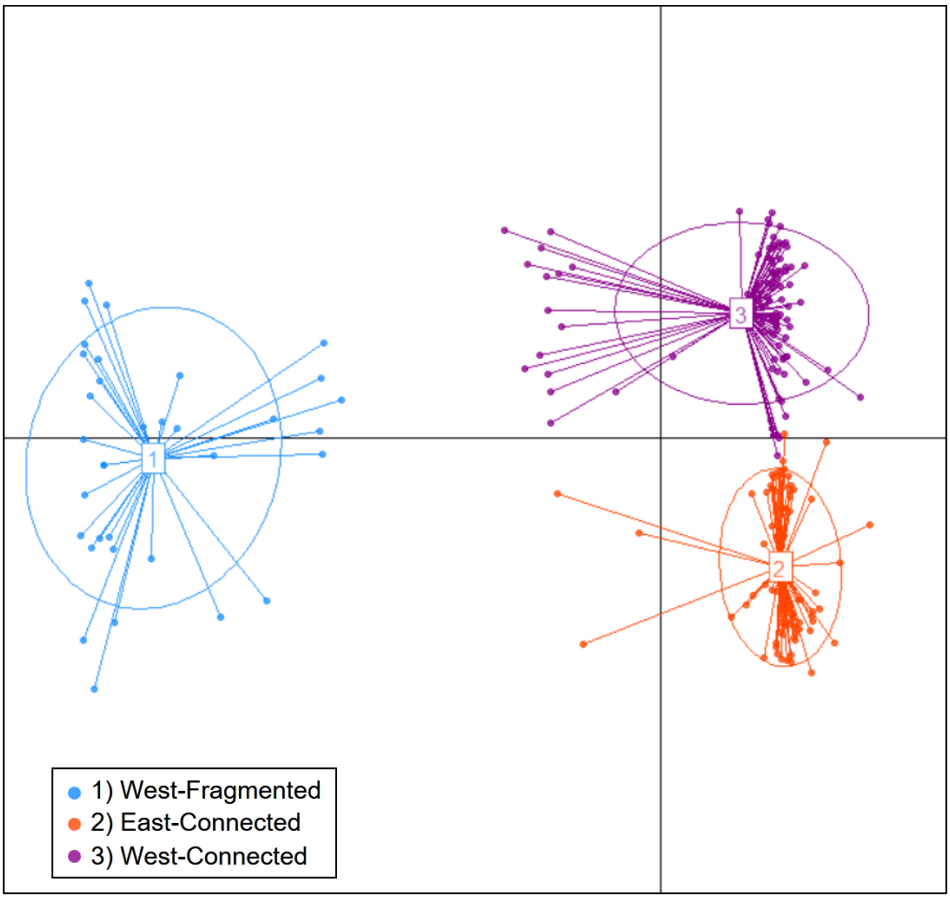

Figure 7: Discriminant analysis of principal component plot demonstrating the genetic distances between the different clusters identified. Group 1 on the chart is the west fragmented population, group 2 is the east connected population, and group 3 is west connected population.

Chapter 5

DISCUSSION

Results of our study indicates an east-west disjunction around the Blue River. General genetic patterns in Texas match those in Oklahoma which indicates that the main stem of the Red River is not a significant barrier to gene flow. In fact, the occurrence of populations near the main stem correlates to an increase in shared genetic information. The cluster identified in the Washita region is demonstrating high amounts of fragmentation (Figure 7) with the groups that are occurring above a dam. Therefore, it appears that the main stem is not the main barrier to movement for the species. Instead, dams and historical patterns seem to be a much larger factor.

The results of the t-test analysis demonstrated that there is a significant decline in genetic diversity above the dams at site OK01, TX04, and TX05. The decreased amount of expected heterozygosity implies a lack of genetic diversity because there are fewer allelic variations present in a population. This could mean there is a lack of new genetic information being brought into the area which could eventually lead to inbreeding depression or speciation. This also strongly suggests that the species is very sensitive to fragmentation. This sensitivity is incredibly important in Texas populations because 40% of all locations where *E. radiosum* were found in Texas were considered fragmented habitats. These results imply a need for stronger management practices to better prevent fragmentation in the future. If the habitat is continuously fragmented this species will not be able to maintain gene flow between populations. Extirpation is a prominent concern in these areas.

The dataset we gathered did not follow expected isolation by distance patterns. This means that there is a more complicated history at play for gene flow than can be explained purely by geographic distances. Most likely, these patterns are caused by dam fragmentation and stream

patterns caused by glacial patterns, which may explain the Blue River speciation. These other causes for fragmentation are more credible for explaining the disjunction of the species than distance alone. Current geographic patterns do little to explain why a completely new species formed in the Blue River. However, glacial patterns tracing back to the Pleistocene more accurately depict how the species could have become so disjunct in habitat. One study by Gehri et al. (2021) explains how river systems are highly complex and can typically be traced back to glacial retreat patterns. These patterns change where a lotic fish is able to survive and move between populations and can even explain why they are found in a specific geographic location. Furthermore, a study on redbreast sunfish (*Lepomis auratus*) demonstrated the distinct separations in populations caused by the presence of dams (Strickland 2022).

Our results suggest that there are three distinct populations of *Etheostoma radiosum*. The west fragmented group shown in Figure 5 as the light blue cluster. The location of these populations above and below the dam indicates that there is likely some amount of shared ancestral lineage still present in the diverged lineages below Lake Texoma in the west. This means that at one point the populations likely had much higher levels of gene flow but have become fragmented by the implementation of dams and other genetic barriers. This can be surmised by the presence of a potential Washita lineage genetic signal that is more predominant in the populations that were sampled above a dam. The purple cluster in Figure 5 shows the sites directly connected to the Red River below lake Texoma and west of the Blue River (west connected). This could mean that there are regional populations that have been confined to small systems below the dam in the area. Further sampling in direct tributaries east and west of the Blue River on the Oklahoma side could better elucidate the range and extent of this group (Figure 8). The orange cluster in Figure 5 indicated the east connected group that has the largest

range size in Oklahoma. It is suggestive that there are more populations that would belong in this lineage in the area highlighted in Figure 8 that my team and I were unable to sample, and this is likely the majority genetic population across all locations of *E. radiosum* as they reach higher occupancy in the Muddy Boggy and Kiamichi system.

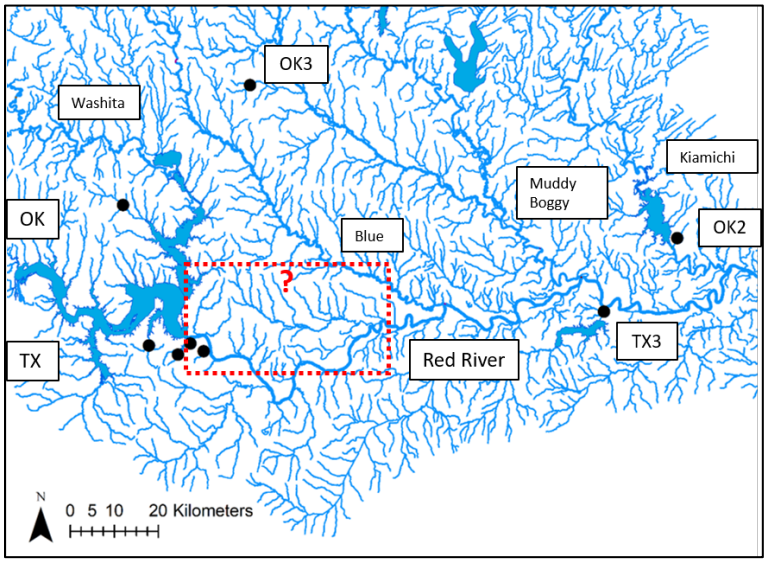

Figure 8: Map of the sites sampled for this project but with an area that was unsampled highlighted to give an understanding of other potential locations to assess genetic patterns for context of a potential west connect genetic cluster. Surveys in Texas in this region yielded no populations, but Oklahoma tributaries may contain some populations.

The analysis done using DAPC demonstrated that these populations do not necessarily follow the genetic patterns you may expect. The two western populations actually had distinctively more genetic distance between them than the west connected had with the east connected population. This indicates that movement through the main stem of the Red River is keeping gene flow prevalent in these genetic clusters. This also means that the dams are having a significant impact on the species and resulting in high amounts of fragmentation. As discussed, this species is showing many signs of being highly sensitive to fragmentation and are therefore

reacting strongly to the presence of the dams. The genetic distance is easily seen in Figure 5 and demonstrates the level to which the dams are disrupting gene flow.

Our hypothesis that there is a distinct split between the populations to the east and west of the Blue River is somewhat supported by this study. However, Texas population 2, Duck Creek, did not fit the pattern. Despite its location it had signals that could have placed it into any of the clusters that were identified during this study. The genetic composition of this particular location gives indications of longitudinal gene flow along the Red River which is not what was anticipated in this area. More importantly, this raises a question of whether the described species by Matthews and Turner (2019) can truly be as isolated as they are described to be if there is such gene flow along the main stem of the Red River. But all populations sampled contained breeding coloration of *E. radiosum palidosum*, further illustrating the strange patterns of the region.

Our results somewhat support the theory put forth by Matthews and Turner (2019) that the Washita lineage (W) is distinct from the eastern lineages for *E. radiosum palidosum*. This species is clearly experiencing some difficulty with gene flow through the main stem of the Red River in Texas. Sites in the Washita system appeared to be isolated from sites in the Muddy Boggy system and the Kiamichi system even though they are relatively close to each other along the main stem. However, the main stem of the Red River between sites in the Muddy Boggy region and Kiamichi region do not show signs of isolation. These sites and the Sanders Creek site (TX03) are all part of the same genetic population. This implies that they are able to use the main stem of the Red River for movement and genetic dispersal in this region. Best management practices would be to avoid any further fragmentation of the system within Texas and maintain and support active populations. *E. radiosum* is an excellent indicator of something that is likely a

larger problem for the fish populations in this area. While the subspecies seem to be stable at the time, continued monitoring of this and other related species will be of high importance in the coming years.

Many studies that investigate the gene flow of darter populations are coming to the conclusion that the fragmented populations are not as distinct as their life history may imply, they should be. One study on Rainbow Darters (*Etheostoma caeruleum*) showed that despite the habitats of the species being heavily fragmented a single genetic population was present in the region (Davis et al. 2014). Most studies suggest that darter species are a spatially limited species that is unable to move large distances. Therefore, the lack of obvious disruptions to gene flow are difficult to explain. Washburn et al. (2020) suggests that perhaps the darters are able to move between populations because of an adaptation that allows dispersal through unstable and patchy habitats. Whereas a study by Luiken et al (2021) explores the possibility that glaciation is the true cause of persisted levels of genetic diversity in these populations. In this study it is suggested that historical glaciation patterns may be just as important to consider as the life history of the species to understand the true genetic patterns that can be observed. This interpretation is bolstered by the results of the study on the impact of contemporary landscape variability on rainbow darter genetic patterns in northeastern Iowa mentioned above (Davis et al. 2014). The authors determined that the only plausible explanation for this lack of genetic distinction was that the species genetic makeup was mostly shaped by glacial retreats and advances throughout the geologic history of the region (Davis et al. 2014).

The manifestation of three genetic populations of *E. radiosum* shows how significant the fragmentation of their habitat is. Ancestral lineages have shown to be an equally powerful force in shaping the amount of gene flow between populations of darters as contemporary habitat

changes. This is likely because the population dynamics of darter species could take tens of thousands of generations to demonstrate the genuine impacts of contemporary fragmentation (Davis et al 2014). The notable lack of genetic diversity above dams in the sites sampled for this study demonstrate a significant problem for the species as a whole. It is likely that the continued fragmentation of this species will continue to cause dramatic changes to their genetic health.

The historical lineages and modern divergences of this species has made it a complicated taxa. The Red River is not demonstrating a direct barrier to gene flow in the species in the east and is potentially showing some signs of genetic exchange in the west as well as shown by the genetic profile of Texas population 2, Duck Creek. What our data is showing is how sensitive this group is to fragmentation especially by dams. While we are seeing that the Texas populations are matching the patterns of drainages in Oklahoma, there are still three distinct lineages in Texas. Since two out of the five Texas populations are located above a dam and are therefore mostly demonstrating a shared ancestral lineage rather than current gene flow between the populations, we can assume persistence of only three lineages is unlikely. Fragmentation will cause the populations to continuously diverge in the future which will cause more genetic distance between these locations.

Future research on this subject is needed. Ideally, more loci should be used to confirm the patterns that we were able to see with the data collected. This will increase the resolution of the genetic clusters, allowing for more definitive groupings. Therefore, if more markers are used it will be easier to understand where the breaks in this population are and therefore easier to determine what the barriers to gene flow are and how we can best solve them. More loci would also make it easier to place sites like TX03 that have signals from all three populations into a definitive grouping. Additional sampling should be done in the area highlighted on Figure 5.

These samples could elucidate our theory of a separate west connected group in the region between Lake Texoma and the Blue River. Other species that are commonly associated with *E. radiosum* could also be analyzed genetically to determine if the darter species is a good signifier for other species or if the other species are able to move through the barriers more effectively.

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VITA

I graduated from the Community College of the Air Force in 2016 with an Associate of Science in criminal justice. In 2020, I graduated from Colorado State University with of Bachelor of Science in fish, wildlife, and conservation biology. I worked with Colorado Parks and Wildlife on multiple occasions. I assisted with fish spawning in spring 2019 at Lake Pueblo State Park. I also worked with PhD candidate Tawni Riepe and Colorado Parks and Wildlife on a study of salmonids. This study included work on cross contamination of whirling disease and the genetic factors contributing to the spread of bacterial kidney disease. This work included genetic isolation and testing, care for endangered species, dissection and processing, blood sampling, fish identification, cross-contamination protocols, and habitat assessments. I also assisted graduate student Carli Baum on the study of environmental components on the reproductive success of *Etheostoma nigrum.* This included fish, egg, and larvae care on a daily basis as well as water quality testing and maintenance. I am currently a graduate research assistant at Texas A&M University Commerce to receive my Master of Science with a focus on biological sciences. I anticipated to graduate in May of 2023. My work will include fisheries biology in a field setting as well as work with genetic material in the laboratory. Other experience in fisheries and biological processes include limnology assessments, streamflow data collections, watershed sciences, biological report writing, fish culturing, and fish physiology.

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